Reflective projective billiards in higher dimensions: monodromy, threshold phenomena, and explicit polytopes

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Abstract

We develop a linear and algebro-geometric framework for reflective projective billiards in projective dimension at least three. Scalar products of harmonic homologies lead to a parity obstruction, a sharp general lower bound for the period, and a trichotomy between the rigid end, the threshold, and the stable range. We construct bounded rational polytopes with primitive open families of periodic trajectories in dimensions three through six, analyze the invariant quadrics and reflection groups of the examples, and reduce the threshold problem in projective three-space to an explicit irreducibility question. We also isolate the determinantal equation behind the rank-three threshold argument and explain exactly why it becomes vacuous in vector rank four.

Keywords. Projective billiards; harmonic homologies; reflection groups; character varieties; periodic trajectories; convex polytopes.

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1 Introduction

Reflective projective billiards turn a dynamical closing condition into a factorisation problem in PGL_{n+1}. This paper studies the dimension-dependent consequences of that observation. The main themes are the number of equations in the scalar-product locus, determinant parity, the rank of partial products, and the geometry needed to convert a word of involutions into a bounded polytope with a strict itinerary.

A projective billiard is *k-reflective* when its billiard map has an open set of k-periodic points; the first examples, of period three and of every even period at least four, are due to Fierobe [4], and Problem 1 of [1] asks which further periods occur. Fierobe's higher-dimensional non-existence theorem for triangular families and his subsequent analysis of caustics provide complementary background for the higher-dimensional setting [5, 6]. The definitions and openness criterion are recalled below, and every construction and auxiliary result used in the higher-dimensional arguments is proved here. For related projective factorisation results, including products of three harmonic homologies, see [7].

The main contributions of this paper are: a dimension-dependent parity and rank framework for scalar monodromy; explicit bounded rational polytopes with primitive open families in dimensions three through six; exact analyses of their invariant quadrics and reflection groups; and a reduction of the threshold question in projective three-space to an explicit irreducibility problem. The irreducibility of V_6 and the resulting threshold theorem are kept explicitly open where the proof does not close.

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2 Setting

A *projective billiard* [8] is a bounded convex polytope $\Omega \subset \mathbb{R}^n$ together with a *transversal line field* assigning to each point in the relative interior of a facet of $\partial\Omega$ a line $L(p)$ through p not contained in the facet hyperplane; reflection is defined by requiring the incoming ray, the outgoing ray, the facet tangent and $L(p)$ to have cross ratio -1 in their impact pencil.

Throughout this paper we restrict to *central facet fields*: on each facet the transversal lines belong to the pencil through one fixed projective centre outside the supporting hyperplane. No reduction from arbitrary transversal fields to this class is asserted. Statements about the period spectrum use the stated convention on the facets visited by the witness.

Definition 2.1 (Fierobe [4])**.** Ω is *k-reflective* if there are a k-periodic orbit $(p_1, \dots, p_k)$ and an **open** $U_1 \times U_2 \subset (\partial\Omega)^2$ containing (p_1, p_2) every point of which begins a k-periodic orbit.

The open set is the whole content of the definition, and it is where the difficulty lives: exhibiting one periodic orbit is easy, exhibiting an open family of initial impact pairs is not.

2.1 Reflection as a harmonic homology

For a covector L and a vector O with $LO \neq 0$ put $H = I - 2\,OL/(LO)$. Then $H^2 = I$, $HO = -O$, $Hx = x$ for $Lx = 0$, and $\det H = -1$.

Proposition 2.2. *Let p lie on L, let $\tau \in \ker L \setminus \{0\}$ be a tangent direction, and put $\nu = O - p$. The involution induced by H on the pencil at p sends $a\tau + b\nu$ to $a\tau - b\nu$; hence the cross ratio of (incoming, outgoing; tangent, transversal) is -1. So reflection in the side with transversal field the pencil through O* is *the harmonic homology with axis the facet hyperplane and centre O.*

Proof. H fixes p and is an involution, so induces an involution of the pencil at p fixing the two distinct lines $\mathbb{R}\tau$ and pO; in the coordinate taking them to 0 and ∞ it is $m \mapsto -m$. Parametrising by $(a : b)$ with direction $a\tau + b\nu$, the tangent is $(1 : 0)$, the transversal $(0 : 1)$, and with $[ij] = a_i b_j - a_j b_i$ the cross ratio of $(a : b), (a : -b), (1 : 0), (0 : 1)$ is $[13][24]/([14][23]) = -1$. □

Consequently, if $H_{j_k} \cdots H_{j_1}$ is scalar then *every* line returns to itself after the corresponding k reflections. The converse is false: an isolated trajectory may well close without the product being scalar, and it is precisely the scalar case that produces an *open set* of closed trajectories, which is what Definition 2.1 demands. If the product is cI, determinants give $c^{n+1} = (-1)^k$. For odd projective dimension n, the left side is positive, so k must be even. For even n, the real scalar is $c = (-1)^k$ and determinants give no additional obstruction. For the central fields considered here, an open family also forces scalar monodromy. The initial lines of a regular family form an open subset of the Grassmannian of projective lines. Their invariance under the fixed product is an algebraic incidence condition; vanishing on an open subset forces it on the entire Grassmannian. A linear map preserving every two-dimensional vector subspace preserves every one-dimensional subspace, obtained as an intersection of such subspaces in vector dimension at least three. Applying this to basis vectors and their pairwise sums shows that the map is scalar. This necessity concerns an open family; an isolated closed orbit does not suffice.

3 An openness criterion

Everything below rests on this; it assumes no symmetry.

Lemma 3.1 (Exit map)**.** *Let Ω be a convex polytope, p in the relative interior of a facet, w pointing into Ω. The ray $p + sw$, $s > 0$, meets $\partial\Omega$ in exactly one point $E(p, w)$, and E is continuous wherever $E(p, w)$ lies in the relative interior of a facet.*

Proof. Ω is convex and bounded, so $s^* = \sup\{s > 0 : p + sw \in \Omega\}$ is finite and $E(p, w) = p + s^*w \in \partial\Omega$; uniqueness is convexity. If $E(p, w)$ is in the relative interior of a facet F then w is not parallel to its supporting hyperplane, so the intersection determinant is nonzero and Cramer's rule gives s^* near (p, w) as a quotient of polynomials with nonvanishing denominator. Since the relative interior of F is open in $\partial\Omega$, nearby data still exit through F. □

Remark 3.2. Requiring nonzero intersection determinants with every supporting hyperplane is unnecessarily restrictive: a strict trajectory may be parallel to facets it never reaches. Continuity only requires transversality at the actual exit facet, together with persistence of the strict inequalities for the other facets.

Theorem 3.3 (Openness criterion)**.** *Let Ω be a convex polytope with facets $F_0, \dots, F_{k-1}$ and transversal field on F_i the pencil through O_i, with $L_iO_i \neq 0$. Assume* (H1) *$H_{k-1} \cdots H_0 = cI$, $c \neq 0$;* (H2) *there is an orbit with itinerary $0, 1, \dots, k-1$ whose impacts are strictly interior to their facets, with strictly positive step parameters, each losing facet excluded by a strict inequality and nonvanishing transversality at each impact;* (H3) *the return direction is $u_k = \kappa u_0$ with $\kappa > 0$. Then Ω is k-reflective; if the k impacts lie on distinct facets the period is primitive.*

Proof. Each step is a line–line intersection followed by a reflection, so the impact parameters and directions are rational functions of the initial data, with denominators nonvanishing at the witness by (H2); the vanishing locus being closed, they are defined and continuous nearby. The finitely many strict inequalities of (H2) persist, and by Lemma 3.1 each of the k exit maps is continuous with value in the relative interior of a facet, which is open in $\partial\Omega$; composing, the itinerary is unchanged for nearby data. By (H1) the product is the identity of PGL_{n+1}, so *every* line returns to itself: $\ell_k = \ell_0$ as an unoriented line, and since ℓ_0 is not the supporting hyperplane of F_0 the intersection $\ell_0 \cap F_0$ is a single point, whence $p_k = p_0$. For the orientation, (H1) only gives $u_k = \kappa u_0$ with κ real nonzero — neither ± 1 nor a priori locally constant — but $\kappa = (u_k \cdot u_0)/(u_0 \cdot u_0)$ is continuous and nonvanishing and equals a positive number at the witness by (H3), so $\kappa > 0$ on a smaller neighbourhood. Finally, a proper period $d \mid k$, $0 < d < k$, would force $p_d = p_0$; but p_d lies on facet F_d and p_0 on F_0, which are distinct facets, and the impacts are interior to them, so $p_d \neq p_0$. □

Hypotheses (H2)–(H3) are finitely many strict inequalities and (H1) one identity, so they can be certified exactly for explicit data. We use this repeatedly below.

4 Higher dimensions

Nothing in the definitions or openness argument used $n = 2$: a harmonic homology of $\mathbb{RP}^n$ is $H = I - 2OL/(LO)$ with L a hyperplane covector, the reflection law of Proposition 2.2 is the same harmonic condition in the pencil of lines at the impact point inside the 2-plane spanned by the ray and the transversal, and the openness criterion of Theorem 3.3 goes through verbatim for a convex polytope of $\mathbb{R}^n$ with impacts interior to facets: its proof uses only convexity, Cramer's rule and finitely many strict inequalities. What changes are the numbers, and they change in a way that explains $k = 4, 5, 6$ as one law.

4.1 The four counts, and a trichotomy

Write $N = \dim \mathrm{PGL}_{n+1} = (n+1)^2 - 1$. A harmonic homology of $\mathbb{RP}^n$ is a pair (L, O) with $LO \neq 0$ modulo two scalings, so $\dim \mathrm{Inv} = 2(n+1) - 2 = 2n$. At a point where the corresponding Jacobian ranks are maximal and the indicated stabiliser is trivial, a direct dimension count gives:

$$\dim V_k = 2nk - N, \qquad \dim(\text{moduli}) = 2nk - 2N = 2n(k-n-2), \qquad \dim S_k = nk + n.$$

For the expected local value of $\dim S_k$: the space of quadrics of $\mathbb{RP}^n$ has dimension $\binom{n+2}{2} - 1$, each mirror is one $n_i \in \mathbb{RP}^n$, and the closure condition is at most $\dim \mathrm{O}(Q) = n(n+1)/2$ conditions, so $\dim S_k = \left[\binom{n+2}{2} - 1\right] + nk - \frac{n(n+1)}{2} = nk + n$.

Proposition 4.1 (Trichotomy of the expected counts)**.** *Under the maximal-rank and stabiliser hypotheses just stated, the expected moduli dimension is* 0 *if and only if* $k = n+2$*, and the expected dimensions of* S_k *and* V_k *agree if and only if* $k = n+3$*; for* $k \geq n+4$*, the expected dimension of* S_k *is smaller. These are the rigid, threshold, and dimension-deficient regimes, respectively.*

Proof. $2n(k-n-2) = 0 \iff k = n+2$. And $2nk - N = nk + n \iff nk = N + n = n^2 + 3n \iff k = n+3$; the inequality for larger k follows since the left side grows twice as fast. □

4.2 Where the counts are valid, and by how much they fail

The counts above are stated at points where the relevant Jacobian has maximal rank. That hypothesis is not automatic, and it fails at three of the examples of these notes. The following two statements say exactly when, and by exactly how much.

Write Φ for the map sending a k-tuple of harmonic homologies to its product in PGL_m, so that $V_k = \Phi^{-1}(\mathrm{id})$, and let $A_j = H_{k-1} \cdots H_{j+1}$ and $H'_j = A_j H_j A_j^{-1}$ be the *unfolded* mirrors, with normals $a_j = A_j n_j$.

Proposition 4.2 (The cokernel)**.** *Under the trace pairing,*

$$\operatorname{coker} d\Phi \;\cong\; \{Y \in \mathfrak{sl}_m : \; [Y, H'_j] = 0 \text{ for all } j\},$$

the traceless part of the commutant of the group generated by the unfolded mirrors. If moreover the mirrors are orthogonal for a positive definite form Q *and the* a_j *span* $\mathbb{R}^m$*, then in coordinates with* $Q = I$ *this is* $\{S \text{ symmetric traceless} : \; Sa_j \parallel a_j \; \forall j\}$*, the space of quadratic forms the configuration preserves beyond the given one.*

Proof. Differentiating $H^2 = I$ gives $\delta H\, H + H\, \delta H = 0$, so $X = \delta H\, H$ satisfies $HXH = -X$; the space $W_H = \{X : HXH = -X\}$ is the -1 eigenspace of the involution Ad_H, of dimension $2(m-1)$, which is the number of parameters of a harmonic homology, and $X \mapsto XH$ is bijective, so the tangent space maps onto W_H. Varying only H_j, $\delta P \cdot P^{-1} = A_j(\delta H_j H_j) A_j^{-1} = W_{H'_j}$, whence $\operatorname{im} d\Phi = \sum_j W_{H'_j}$. Now $\mathrm{Ad}_{H'}$ is self-adjoint for the trace form, so $W_{H'}^{\perp}$ is its $+1$ eigenspace, the centraliser of H'; and the dual of $\mathfrak{pgl}_m$ under that pairing is $\mathfrak{sl}_m$. For the last sentence, $[Y, I - 2aa^{\top}/|a|^2] = 0$ forces $Ya = \lambda a$ and $Y^{\top} a = \lambda a$; splitting Y into symmetric and antisymmetric parts and subtracting gives $Ka = 0$ for the antisymmetric part, hence $K = 0$ when the a_j span. □

Proposition 4.3 (The corank is the number of blocks, minus one)**.** *Suppose the mirrors are orthogonal for a positive definite form, their normals span* $\mathbb{R}^m$*, and their representation*

decomposes into s pairwise non-isomorphic irreducible summands. Then $d\Phi$ has corank $s-1$, and the tangent space to the scalar-product equations has dimension

$$\dim\ker d\Phi = 2kn - \big((n+1)^2 - 1\big) + (s-1).$$

For $s = 1$ the map is a submersion and this is also the local dimension of V_k. For $s > 1$ it is only a tangent-space calculation; higher-order equations may decrease the local dimension.

Proof. The unfolded mirrors generate the same group as the original ones, since $H'_{k-1} = H_{k-1}$ and each H_j is recovered inductively from the H'. By Proposition 4.2 the cokernel is the traceless part of the commutant. By the positive definite and spanning hypotheses in Proposition 4.2, every commuting endomorphism is symmetric. The eigenspaces of a symmetric commuting endomorphism are invariant, so on each irreducible block it is scalar. There are no off-diagonal intertwiners between non-isomorphic blocks. The commutant therefore has dimension s, and its traceless part has dimension $s-1$. Rank–nullity gives the formula; the submersion theorem applies when $s = 1$. □

Proposition 4.4 (A quadratic obstruction at the rank-five example)**.** *At the configuration of Theorem 4.25, the tangent space to the scalar-product equations has dimension* 33*, whereas the complex local dimension is* 32*. The real local dimension is at most* 32*.*

Proof. Use independent affine coordinates for the seven poles p_i and axes l_i, fixing the first nonzero coordinate of each at its value in the displayed root n_i. There are $7 \cdot 8 = 56$ coordinates. For $H_i = I - 2p_i l_i/(l_i p_i)$ and $P = H_6 \cdots H_0$, exact differentiation at $p_i = n_i$, $l_i = n_i^\top$ gives $P = -I$ and $\operatorname{rank} dP = 23$. The matrix $Y = \operatorname{diag}(1, 1, 1, -4, 1)$ commutes with every mirror, and $\operatorname{tr}(Y\, dP) = 0$ for all 56 coordinate directions.

Here is an explicit tangent direction, with indices starting at zero; all unlisted coordinate derivatives are zero:

coordinate	$p_{0,3}$	$l_{0,3}$	$l_{1,3}$	$p_{2,1}$	$p_{2,4}$	$l_{2,1}$	$l_{2,2}$
derivative	2	2	2	−1	1	−1	1

Along the straight coordinate path tv, rational multiplication gives

$$P'(0) = 0, \qquad \operatorname{tr}(Y(P(t) + I)) = \frac{2t^2(18t^2 - 5)}{(t^2+1)(2t^2+1)} = -10t^2 + O(t^4).$$

The supplement's verifier `check_corank_obstruction.py` reproduces these identities and the rank in exact arithmetic.

Eliminate 23 independent equations by the analytic implicit function theorem. Their zero set is smooth of dimension 33 and has tangent space $\ker dP$. A curve in that set with initial derivative v may change the second derivative of P by an element of $\operatorname{im} dP$ only. Pairing with Y annihilates that change. Thus the remaining scalar equation has a nonzero quadratic term, and the local zero set has dimension at most 32. Finally P takes values in the smooth determinant--1 hypersurface of dimension 24, so its fibre is locally defined by 24 equations in 56 variables. The complex Krull bound is at least 32 on every component. The complex local dimension is consequently exactly 32. □

Remark 4.5 (Which of our examples are which)*.* Of the five higher-dimensional configurations certified in this paper, two are irreducible and three are not; the split is exactly the one predicted by Theorem 4.31: the reducible ones are precisely the words of B_m with isolated

coordinates.

configuration	s	corank	counts valid?
$\mathbb{R}^3,\ k=6$ (Thm. 4.13)	1	0	yes
$\mathbb{R}^3,\ k=6$ (Thm. 4.23)	1	0	yes
$\mathbb{R}^4,\ k=7$ (Thm. 4.25)	2	1	no
$\mathbb{R}^5,\ k=8$ (Thm. 4.27)	3	2	no
$\mathbb{R}^6,\ k=9$ (Thm. 4.29)	4	3	no

The block structure is the same in the last three: one D_4 block of rank four, plus one line for each isolated coordinate. The three lines carry pairwise different characters of the group — on each, one generator acts by -1 and the others trivially — so they are non-isomorphic and Proposition 4.3 applies. Checked over $\mathbb{Q}$ by `verificadores/bloques.py` and `verificadores/tolerancias.py`.

None of this disturbs the examples themselves: they are certified by the openness criterion, which uses no dimension count. What it disturbs is the right to apply the counts, and Proposition 4.1 with them, at those points.

4.3 Parity: the obstruction reverses

When n is even, $-I$ is a scalar in odd vector rank, so odd k carries no determinant obstruction. In odd projective dimensions the parity restriction is different, as the theorem shows.

Theorem 4.6 (Parity)**.** *A harmonic homology of $\mathbb{RP}^n$ has a canonical lift $H = I - 2OL/(LO)$, which fixes its axis pointwise and satisfies $H^2 = I$ and $\det H = -1$. When $n+1$ is even, this determinant is independent of the choice between the two involutive lifts H and $-H$. If the canonical lifts satisfy $H_{k-1}\cdots H_0 = cI$ then*

$$c^{n+1} = (-1)^k.$$

*Hence for n even there is no parity constraint on k — which is why $\mathbb{RP}^2$ admits odd k — while for n **odd, k must be even**. In particular in $\mathbb{RP}^3$ there is no k-reflective projective billiard with k odd; the rigid case $k = n+2 = 5$ is* empty*, not merely degenerate.*

Proof. Take determinants. For $n+1$ even, $\det(-H) = (-1)^{n+1}\det H = \det H$, so the sign of the lift does not matter. If $n+1$ is even then $c^{n+1} > 0$ for real $c \neq 0$, forcing $(-1)^k > 0$. □

Remark 4.7 (The quiver picture, and where the parity actually sits). It is worth setting Theorem 4.6 inside the Deligne–Simpson machinery, because the same set-up is used in the next paragraph. Rescale all the factors by one μ so that the product is the identity of GL_{n+1}: the local classes become μ with multiplicity n and $-\mu$ with multiplicity 1, the associated star-shaped quiver of [3] has central dimension $n+1$ and k legs of dimension $\operatorname{rank}(A_i - \xi_{i1}) = 1$, so $\alpha = (n+1; 1, \dots, 1)$, and Crawley-Boevey's necessary condition reads

$$\xi[\alpha] = \prod_{i=1}^{k}\prod_{j}\xi_{ij}^{m_{i,j-1}-m_{ij}} = \prod_{i=1}^{k}\mu^{n}(-\mu) = (-1)^k\mu^{k(n+1)} = 1.$$

This is precisely $\prod_i \det A_i = 1$; over $\mathbb{C}$ it can always be arranged by the choice of μ, so it is not an obstruction there. The parity is a statement about the *real* problem: μ must be real, and when $n+1$ is even $\mu^{k(n+1)} > 0$, forcing $(-1)^k = 1$. So the quiver normalisation recovers Theorem 4.6, but by the same determinant identity rather than by an independent route.

The same computation verifies the other two conditions of [3, Thm. 10] for $n = 2$, $k = 5$: $\alpha = (3; 1^5)$ is a positive imaginary root, and $p(\alpha) = 2$ exceeds the largest value 1 attained by

$p(\beta)+p(\gamma)+\cdots$ over the fifteen nontrivial decompositions of α into positive roots with $\xi = 1$. So there exists a solution that is *irreducible as a representation*: five harmonic homologies with scalar product whose mirrors have no common invariant subspace, for every k satisfying the parity condition, with no construction needed. That is not the same as irreducibility of the *variety* V_k, which is what Theorem 4.17 needs, and the criterion says nothing about it.

4.4 The rigid end, in every dimension

The low-dimensional case needed in the argument is the following elementary fact.

Lemma 4.8 (Four reflections in dimension three)**.** *Let R_0, R_1, R_2, R_3 be orthogonal reflections of a positive definite three-dimensional space and suppose $R_0R_1R_2R_3$ is scalar. Then their normals span a subspace of dimension at most two.*

Proof. The product has determinant $+1$, so the scalar is I. Thus $R_0R_1 = R_3R_2$. If this product is the identity, each pair consists of the same reflection and the four normals span at most a two-dimensional space. Otherwise the common product is a nonidentity rotation with a unique fixed axis. The axis of a product of two reflections is perpendicular to both normals, so all four normals lie in the perpendicular plane. □

Theorem 4.9 ($k = n+2$ is always degenerate)**.** *Let Q be positive definite on $\mathbb{R}^m$, $m = n+1 \geq 3$, let $R_0, \ldots, R_{k-1}$ be Q-reflections with $k = n+2 = m+1$, and suppose $R_0R_1 \cdots R_{k-1} = cI$. Then $c = 1$, m is odd, and the normals $n_0, \ldots, n_{k-1}$ do **not** span $\mathbb{R}^m$. Consequently all k mirrors contain a common projective point, and the region they bound is unbounded: there is no such billiard.*

Proof. The product lies in $\mathrm{O}(Q)$, and $cI \in \mathrm{O}(Q)$ forces $c = \pm 1$. Taking determinants, $c^m = (-1)^k = (-1)^{m+1}$; with $c = -1$ this reads $(-1)^m = (-1)^{m+1}$, impossible, so $c = 1$ and m is odd. (For m even the statement is vacuous by Theorem 4.6.)

Suppose the normals span $\mathbb{R}^m$. Put

$$V = \langle n_0, n_1, n_2 \rangle, \quad W = \langle n_3, \ldots, n_{k-1} \rangle, \quad X = R_0R_1R_2, \quad Y = R_3 \cdots R_{k-1},$$

so $XY = I$ and $X = Y^{-1}$; note $k - 3 = m - 2 \geq 1$, so W is defined. Each R_i with $i \leq 2$ fixes $n_i^{\perp}$ pointwise, so X fixes $V^{\perp}$ pointwise; likewise Y^{-1} fixes $W^{\perp}$ pointwise. Hence X fixes $V^{\perp} + W^{\perp} = (V \cap W)^{\perp}$ pointwise. Since $V + W = \mathbb{R}^m$,

$$\dim(V \cap W) = \dim V + \dim W - m \leq 3 + (m-2) - m = 1,$$

so $\operatorname{codim} \operatorname{Fix}(X) \leq 1$. As $\det X = (-1)^3 = -1$ we have $X \neq I$, so $\operatorname{Fix}(X)$ is exactly a hyperplane; an element of $\mathrm{O}(Q)$ fixing a hyperplane pointwise and not the identity acts by -1 on the orthogonal line, so $X = R'$ is a *reflection*, with normal n' spanning $\operatorname{Fix}(X)^{\perp}$. From $\operatorname{Fix}(X) \supseteq (V \cap W)^{\perp}$ we get $n' \in V \cap W$, so $\dim(V \cap W) = 1$ and the inequality above is an equality: $\dim V = 3$.

Now $R_0R_1R_2R' = I$ with n_0, n_1, n_2, n' all in V, which is 3-dimensional and on which Q is definite; each of the four maps preserves V and restricts to a reflection of it. By Lemma 4.8 the four normals are coplanar, so $\dim V \leq 2$ — contradicting $\dim V = 3$.

Finally, if the normals span a proper subspace U then every mirror contains $\mathbb{P}(U^{\perp}) \neq \emptyset$. Pick a point p in it: if p is affine the region is a cone with apex p, and if p is at infinity with direction d every mirror is invariant under translation by d and so is the region. Either way it is unbounded. □

Remark 4.10 (Arbitrary signature). Definiteness of Q was used in exactly two places, and both can be replaced by an explicit hypothesis, so the theorem holds over any nondegenerate Q — in particular over $\mathbb{C}$, where all nondegenerate forms are equivalent:

(a) $\mathbb{R}^m = V \oplus V^\perp$ with each R_i preserving both, which needs $V = \langle n_0, n_1, n_2 \rangle$ **nondegenerate**;

(b) "an element of $\mathrm{O}(Q)$ fixing a hyperplane pointwise and not the identity acts by -1 on the orthogonal line", which needs $\mathrm{Fix}(X)$ **nondegenerate**, i.e. its orthogonal line non-isotropic.

Everything else is signature-free: the identity $(U_1 \cap U_2)^\perp = U_1^\perp + U_2^\perp$ holds for any nondegenerate Q, and $\mathrm{Fix}(R_1R_0) = n_0^\perp \cap n_1^\perp$ exactly whenever n_0, n_1 are independent, since $R_1R_0x = x$ forces $\frac{\langle x,n_0\rangle}{\langle n_0,n_0\rangle} n_0 = \frac{\langle x,n_1\rangle}{\langle n_1,n_1\rangle} n_1$ and hence both coefficients to vanish — no appeal to signature anywhere. Moreover the relation is invariant under cyclic rotation of the mirrors, since rotating conjugates the product, so the split may be taken at any of the k positions and it suffices that *some* three cyclically consecutive normals span a nondegenerate 3-space. Whether all k rotations can fail simultaneously we do not determine. The billiard application below uses a positive definite Q, so these signature issues do not arise.

Remark 4.11. Only the case $j = 3$ of the splitting is used, and that is what reduces the argument to Lemma 4.8: three reflections have determinant -1, which is what rules out $X = I$ and forces $\mathrm{Fix}(X)$ to be a genuine hyperplane. A split into two even halves would allow $X = I$ and prove nothing.

4.5 The threshold: what is proved, and what is not

At $k = n + 3$ the expected dimensions coincide. The following explicit example illustrates the threshold case in dimension three.

Proposition 4.12 ($\mathbb{RP}^3$, $k = 6$)**.** *There is an explicit configuration of six Q-reflections of $\mathbb{R}^4$ with integer normals,*

$$n_0 = (1, 0, 0, -3), \quad n_1 = (2, 1, -1, 1), \quad n_2 = (1, 3, 2, 0),$$
$$n_3 = (0, -1, 4, 0), \quad n_4 = (0, -55, 50, 17), \quad n_5 = (-171, 215, -40, -82),$$

with $Q = x_0^2 + x_1^2 + x_2^2 + x_3^2$ and $R_5R_4R_3R_2R_1R_0 = -I$ exactly over $\mathbb{Q}$, whose normals have rank 4, at which

$$\dim V_6 = 6k - 15 = 21, \qquad \dim S_6 = 3k + 3 = 21, \qquad \dim(\textit{moduli}) = 6k - 30 = 6,$$

these being the values on the component through the point. Consequently S_6 contains the irreducible component of V_6 through that point: ***every configuration in that component preserves a quadric****. The invariant form is nondegenerate on a nonempty open subset of this component, since it is positive definite at the displayed point.*

Proof. The three Jacobian ranks and the rank of the linear system for the invariant quadric are computed over $\mathbb{Q}$ by Gaussian elimination at the displayed point: Jacobian of the 15 equations has rank 15, the infinitesimal stabiliser is spanned by I, the Jacobian in the n_i directions has rank $6 = \dim \mathrm{O}(Q)$, and the 24×10 system for Q has rank 9, so the invariant quadric is unique up to scale. Semicontinuity propagates each value to the component. Since $\dim S_6 = \dim V_6$ and $S_6 \subseteq V_6$ is closed, S_6 is a union of components of V_6, and it contains the given point. □

4.6 A 6-reflective projective billiard in $\mathbb{R}^3$

The counts and Proposition 4.12 say where to look. Here is an actual example. Its priority relative to the literature is not asserted here.

Theorem 4.13. *Let $Q = x_0^2 + x_1^2 + x_2^2 + x_3^2$, let $n_i = (a_i, d_i)$ be the six integer vectors*

$$n_0 = (-1, 3, -4, -1), \quad n_1 = (-1, 0, 4, 1), \quad n_2 = (3, 4, -1, -2),$$
$$n_3 = (0, -8, 91, 0), \quad n_4 = (0, 2198, 2119, 8345), \quad n_5 = (-1577, 276, 1117, -146),$$

let $\varepsilon = (+, +, +, +, -, +)$, and let

$$\Omega = \{x \in \mathbb{R}^3 : \ \varepsilon_i(a_i \cdot x + d_i) < 0, \ i = 0, \dots, 5\},$$

*with the transversal field on facet i the pencil of lines through the pole $O_i = n_i$ of that facet. Then Ω is a **bounded** convex polytope and the projective billiard on it is 6-**reflective**, with primitive period 6.*

Proof. The hypotheses of Theorem 3.3, whose proof applies verbatim to a convex polytope of $\mathbb{R}^n$ with impacts interior to facets, are verified in exact rational arithmetic.

(H1). Each $R_i = I - 2n_i n_i^\top/(n_i \cdot n_i)$ satisfies $R_i^2 = I$, $\operatorname{tr} R_i = 2$ and $R_i^\top R_i = I$, and $R_0R_1R_2R_3R_4R_5 = -I$ over $\mathbb{Q}$, computed in both orders. The six normals have rank 4 and the six facets are pairwise distinct.

Boundedness. The recession cone is $\{y : \varepsilon_i a_i \cdot y \le 0 \ \forall i\}$, and it is $\{0\}$ exactly when the inward normals positively span $\mathbb{R}^3$, i.e. when they have rank 3 and some $\mu \in \mathbb{Q}^6_{>0}$ satisfies $\sum_i \mu_i \varepsilon_i a_i = 0$. Both hold, with $\mu = (\frac{8056}{65}, \frac{5581}{5}, \frac{61038}{65}, 1, 2, 1)$ and the identity checked exactly. Nonemptiness: $x_0 = (-\frac{49}{100}, -\frac{153}{100}, -\frac{121}{100})$ satisfies all six inequalities strictly.

(H2). The witness orbit starts at $p_0 = (-\frac{1}{2}, -\frac{3}{2}, -\frac{5}{4})$, which lies on facet 0 and strictly inside the other five, with direction $u_0 = (-\frac{4}{9}, -\frac{7}{10}, \frac{5}{9})$ pointing inwards. Its itinerary is $5, 4, 3, 2, 1, 0$; at each of the six steps the first hit (least step parameter) is the predicted facet, with $s > 0$ strictly, the impact strictly interior to that facet, every losing facet excluded by a strict inequality, no non-winning facet parallel to the ray, and nonvanishing transversality.

(H3). After six steps $p_6 = p_0$ and $u_6 = u_0$ exactly, so $\kappa = 1 > 0$. The six impacts lie on six distinct facets, so the period is primitive.

Reflection is implemented projectively: at an impact q on facet i the outgoing direction is $w - w_4 q$ where $(w, w_4) = R_i\,(u, 0)^\top$, which is the harmonic condition of Proposition 2.2 in the 2-plane spanned by the ray and the transversal. □

Proposition 4.14 (The explicit $\mathbb{R}^3$ billiard is not isolated)**.** *In the space V_6 of six harmonic homologies with scalar product, the configuration of Theorem 4.13 is a smooth real point of dimension 21. Moreover, there is a relative open neighbourhood of it in V_6 all of whose configurations define bounded convex 6-reflective projective billiards with the same primitive itinerary. Its local moduli space modulo PGL_4 has dimension 6.*

Proof. Use independent affine coordinates for the axes and centres of the six homologies. The exact Jacobian of the 15 traceless scalar-product equations at the displayed rational configuration has rank 15; hence the implicit-function theorem gives a smooth local $36-15 = 21$ dimensional locus. The infinitesimal stabiliser has dimension 1 in $\mathfrak{gl}_4$, consisting only of scalar matrices, so the projective orbit has dimension 15 and the local quotient has dimension 6.

All the remaining conditions in the proof of Theorem 4.13 are strict: a positive dependence among the inward normals, a strictly interior point, strict first-hit inequalities, interior impacts, nonvanishing transversality, and positive return multiplier. They persist under sufficiently small perturbations. The product relation supplies the closing line, and the same continuity argument as in Theorem 3.3 supplies the oriented return. Thus the openness criterion applies throughout a smaller neighbourhood. The rank computation uses the same exact rational Jacobian calculation as the verifier for Proposition 4.12, now evaluated at the billiard configuration itself. □

Remark 4.15. This lands exactly at the threshold: $k = 6 = n+3$ for $n = 3$. And k is even, as Theorem 4.6 requires for odd n — there is no 5-reflective example in $\mathbb{R}^3$ to be found, which is why the rigid case is empty rather than merely degenerate. Two things are *not* claimed. This configuration and the one of Proposition 4.12 are different, and we do not claim they lie on the same component of V_6; and irreducibility of V_6 in dimension 3 is open, as is the threshold theorem itself (Corollary 4.18). What is proved here is existence.

Finally, this example is not a kaleidoscope in disguise: its normals are rational, so each $\operatorname{tr}(R_aR_b) = 2 + 2\cos\phi_{ab}$ is rational, and by Niven's theorem all fifteen values of $\cos\phi_{ab}$ fall outside $\{0, \pm\frac{1}{2}, \pm 1\}$. So every one of the fifteen pairs generates an infinite dihedral group and the mirror group is infinite.

The configuration is produced by a factorisation argument with the splitting $6 = 3 + 3$ and an elementary property of $\mathrm{O}(4)$:

Lemma 4.16. *Every element of* $\mathrm{O}(4)$ *of determinant* -1 *has eigenvalue* 1*, hence is a product of at most three reflections.*

Proof. In the block normal form the number of (-1)-blocks is odd, so 1 or 3; in both cases $4 = 2a + \#\{\pm 1 \text{ blocks}\}$ forces at least one $(+1)$-block. □

Choose rational normals n_0, n_1, n_2 and set $X = -(R_2R_1R_0)^{-1} \in \mathrm{O}(4)$, so $\det X = -1$. By Lemma 4.16, X fixes a nonzero rational vector. Choose a rational v with $Xv \neq v$ and put $u = Xv - v$. The reflection R_u interchanges v and Xv, so $Y = R_uX \in \mathrm{SO}(4)$ fixes v and restricts to an element of $\mathrm{SO}(v^\perp) \cong \mathrm{SO}(3)$. If this restriction is not the identity, let e span its fixed axis and choose a rational $a \in v^\perp \cap e^\perp$. The reflection R_a fixes e and reverses the orientation of $e^\perp$, so $R_aYR_a = Y^{-1}$. Put $R_b := YR_a$. Then $R_b^2 = I$, it fixes e, and its determinant on $v^\perp$ is -1, so its other eigenvalues are $+1$ and -1; hence R_b is a reflection and $Y = R_bR_a$. If the restriction is the identity, choose any rational reflection R_a of $v^\perp$ and put $R_b = R_a$. Thus $Y = R_bR_a$, and

$$R_uR_bR_aR_2R_1R_0 = -I.$$

All choices and operations are rational, giving the claimed construction.

4.7 The threshold reduces to irreducibility

The following dimension argument isolates the additional hypothesis needed to turn the threshold count into a theorem: irreducibility of the scalar-product locus.

Theorem 4.17 (The threshold reduces to irreducibility)**.** *Fix* $n \geq 2$ *and* $k = n + 3$*, and suppose there is a point of* $V_k \subset \mathbb{RP}^n$ *at which*

(a) *the Jacobian of the* $N = (n+1)^2 - 1$ *scalar-product equations has rank* N*, so that the component through it has dimension* $2nk - N$*; and*

(b) *the component of the locus* S_k *of configurations preserving a nonzero quadratic form through this point has dimension* $nk + n$*.*

If V_k *is irreducible, then*

$$V_{n+3} = S_{n+3}:$$

every *configuration of* $n + 3$ *harmonic homologies of* PGL_{n+1} *with scalar product preserves a nonzero quadric.*

Proof. By (a) and irreducibility, $\dim V_k = 2nk - N$. By Proposition 4.1, $k = n + 3$ gives $nk + n = 2nk - N$, so by (b) $\dim S_k \geq \dim V_k$. But S_k is a closed subvariety of the irreducible V_k, and a closed subvariety of an irreducible variety of the same dimension is the whole variety. □

For $n = 3$ the hypotheses (a) and (b) are the content of Proposition 4.12, certified in exact arithmetic at the displayed rational point. So:

Corollary 4.18. *If $V_6 \subset \mathbb{RP}^3$ is irreducible, then every configuration of six harmonic homologies of* PGL_4 *with scalar product preserves a quadric — the threshold theorem in dimension three.*

This is worth stating because it moves the problem to where there are tools. By the dictionary of Section 4, irreducibility of V_k is irreducibility of a relative character variety of the k-punctured sphere with involution monodromies, which is the multiplicative Deligne–Simpson problem for the star-shaped quiver with central dimension $n + 1$ and k legs of dimension 1.

4.8 Why the grouping argument could not have generalised

The pair grouping gives a useful dimension count, but it cannot by itself settle the threshold in higher dimensions. The pair lemma is as follows.

Lemma 4.19 (Pair lemma in dimension m)**.** *Let H_0, H_1 be harmonic homologies of $\mathbb{RP}^{m-1}$ with axes $L_0 \neq L_1$ and let $A = H_1H_0$. Then A fixes $L_0 \cap L_1$ pointwise and, if A is regular in the sense that its eigenvalues are $\lambda, \lambda^{-1}, 1, \dots, 1$ with $\lambda \neq \pm 1$ and the 1-eigenspace is exactly $L_0 \cap L_1$, the quadrics preserved by A are exactly those preserved by* both *H_0 and H_1, and they form a linear space of dimension*

$$1 + \binom{m-1}{2}.$$

A single harmonic homology preserves a linear space of quadrics of dimension $1 + \binom{m}{2}$.

Proof. Both H_i fix $L_0 \cap L_1$ pointwise, hence so does A; and $H_0AH_0 = A^{-1}$, so H_0 interchanges the λ- and λ^{-1}-eigenlines and preserves the 1-eigenspace. In a basis $(e_\lambda, e_{\lambda^{-1}}, f_3, \dots, f_m)$ adapted to this, $\tau_A(Q)$ multiplies Q_{ij} by $\lambda_i\lambda_j$, so Q is A-invariant exactly when $Q_{ij} = 0$ unless $\lambda_i\lambda_j = 1$: this leaves Q_{12} and the whole block on the 1-eigenspace, of dimension $1+\binom{m-1}{2}$. In the same basis $H_0 = \begin{pmatrix} 0 & q & \\ r & 0 & \\ & & I \end{pmatrix}$ with $qr = 1$, since $H_0^2 = I$ and H_0 is the identity on $L_0 \supseteq L_0 \cap L_1$. For $Q = 2\beta x_1x_2 + Q'$ with Q' on the 1-eigenspace, $Q(H_0x) = 2\beta qr\, x_1x_2 + Q' = Q$. So every A-invariant quadric is H_0-invariant, hence also $H_1 = AH_0$-invariant; the converse is immediate. For a single homology, in a basis where $H = \mathrm{diag}(-1, 1, \dots, 1)$ the invariant quadrics are those with $Q_{1j} = 0$ for $j \geq 2$: dimension $1 + \binom{m}{2}$. □

Proposition 4.20 (The deficit)**.** *Inside the projective space of quadrics of $\mathbb{RP}^{m-1}$, of dimension $\binom{m+1}{2} - 1$, the locus $\mathcal{Q}_A$ of Lemma 4.19 has codimension $2m - 2$. Hence p pairs of mirrors can be expected to share a quadric only when*

$$p\,(2m-2) \leq \binom{m+1}{2} - 1, \qquad \textit{i.e.} \qquad p \leq \frac{m+2}{4}.$$

At the threshold $k = n+3 = m+2$ the grouping uses $p = \lfloor (m+2)/2 \rfloor$ pairs, and the shortfall $p(2m-2) - \left(\binom{m+1}{2} - 1\right)$ is

m	3	4	5	6	7	8
deficit	3	9	10	20	21	35

Proof. $\binom{m+1}{2} - 1 - \binom{m-1}{2} = 2m - 2$, and the rest is arithmetic. □

Remark 4.21. At $m = 3$ the tabulated deficit is 3, but the grouping there is $2 + 2 + 1$, not $2 + 2$. The fifth homology is $-N^{-1}$, so the real requirement is that one pencil $\mathcal{P}_A$ meet the polarity condition for N, which is $m - 1 = 2$ conditions on a 1-parameter pencil: a shortfall of exactly **1**. And the relation pays exactly one condition, namely $\det(B - I) = 0$, which holds automatically because $B = H_3H_2$ is a product of two involutions. That coincidence — one condition needed, one condition supplied — is a special feature of $m = 3$. For $m \geq 4$ the shortfall is 9 or more while the relation still supplies one condition. So the grouping argument does not merely resist generalisation: it cannot generalise. Theorem 4.17 is the way through, and it needs irreducibility instead.

Remark 4.22 (What irreducibility of V_6 would take). One can try a fibration over the first three factors. Split $6 = 3 + 3$: the relation reads $H_5H_4H_3 = c(H_2H_1H_0)^{-1} =: X$, and $H_5X = H_4H_3$ must be a product of *two* harmonic homologies, which by Lemma 4.19 means $\operatorname{rank}(H_5X - I) \leq 2$.

Now comes the point. Writing $H_5 = I - 2w\tilde{w}/s$,

$$H_5X - I = (X - I) - \tfrac{2}{s}\, w\,(\tilde{w}X),$$

a **rank-one perturbation** of $X - I$. A rank-one perturbation lowers rank by at most one, so $\operatorname{rank}(H_5X - I) \leq 2$ *forces* $\operatorname{rank}(X - I) \leq 3$, i.e. $\det(X - I) = 0$: X must have eigenvalue 1. For general harmonic homologies that is **not** automatic — a product of three of them has determinant -1 but need not fix a point — and it is one condition. Given it, $\operatorname{rank}(X - I) = 3$ and the three further conditions for a rank-one perturbation of a rank-3 matrix to have rank 2 are that w lie in $\operatorname{im}(X - I)$, that $(\tilde{w}X)^\top$ lie in $\operatorname{im}\big((X - I)^\top\big)$, and one scalar normalisation. So the count is

$$\underbrace{18 - 1}_{H_0,H_1,H_2 \text{ with } \det(X-I)=0} + \underbrace{6 - 3}_{H_5 \text{ in the determinantal locus}} + \underbrace{1}_{\text{factorising } H_4H_3}$$
$$= 17 + 3 + 1 = 21 = \dim V_6.$$

And the asymmetry that appears here is exactly the reason S_6 can be a *component* rather than a thin subvariety. On S_6 the mirrors are Q-reflections, so $X \in \mathrm{O}(Q)$ with $\det X = -1$, and then Lemma 4.16 makes $\det(X - I) = 0$ **automatic**: the $3 + 3$ split is free there and costs nothing, while off S_6 it costs a condition. That is where the excess dimension comes from.

What remains, and it is now a question about two explicit varieties rather than about billiards: is the hypersurface $\{\det(X - I) = 0\} \subset \mathrm{Inv}^3$ irreducible, and is the determinantal locus of admissible H_5 irreducible of dimension 3 over it? With both, V_6 is irreducible and Corollary 4.18 gives the threshold theorem in dimension three.

Numerical evidence points that way: Newton's method from sixteen random starts converged to sixteen points of V_6, and *every one* of them preserved a quadric, with the smallest singular value of the linear system at the level of machine epsilon while the next one was of order 10^{-1}. That is evidence, not proof, and it is recorded as such.

4.9 A second example in $\mathbb{R}^3$, and the dichotomy repeats

The example of Theorem 4.13 is not a kaleidoscope. There is also one with data in $\{0, \pm 1\}$, whose mirror group is finite.

Theorem 4.23. *Let $Q = x_0^2 + x_1^2 + x_2^2 + x_3^2$ and take the six mirrors with normals*

$$n_0 = (1, 0, 1, 0), \quad n_1 = (1, 0, -1, 0), \quad n_2 = (1, 0, 0, -1),$$
$$n_3 = (0, 1, 1, 0), \quad n_4 = (0, 1, -1, 0), \quad n_5 = (0, 1, 0, -1),$$

*which are six of the twelve mirrors of the reflection group D_4, with $\varepsilon = (-,-,+,-,-,+)$ and $\Omega = \{x : \varepsilon_i(a_i \cdot x + d_i) < 0\}$. Then Ω is a bounded convex polytope and the projective billiard on it is 6-reflective with primitive period 6; the witness orbit starts at $p_0 = (\frac{4}{7}, \frac{4}{5}, -\frac{4}{7})$ with direction $u_0 = (15, 5, 3)$, has itinerary $2, 1, 3, 5, 4, 0$, and closes with $p_6 = p_0$ and $\kappa = 1$ exactly over $\mathbb{Q}$. Its mirror group is **finite**, of order 96 in PGL_4 and 192 in $\mathrm{O}(4)$: it is a spherical kaleidoscope of $\mathbb{RP}^3$.*

Proof. As in Theorem 4.13, all hypotheses of Theorem 3.3 are certified in exact rational arithmetic: the six maps are Q-reflections whose product in the order of the itinerary is $\pm I$; the recession cone is $\{0\}$, by rank 3 of the inward normals together with an explicit $\mu \in \mathbb{Q}^6_{>0}$ with $\sum \mu_i \varepsilon_i a_i = 0$; the point $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{4})$ is interior; and at each of the six steps the first hit is the predicted facet, strictly interior to it, with every losing facet excluded strictly. Finiteness of the group is a breadth-first closure in PGL_4, exact. □

Remark 4.24. In dimension three these examples exhibit both behaviours: Theorem 4.23 has a finite reflection group, while Theorem 4.13 has an infinite mirror group. The latter shows that the moduli space is not a list of symmetries.

A remark on how it was found, because it is a methodological point. A first sweep over the mirrors of D_4, B_4 and F_4 found nothing, because it fixed $\varepsilon_0 = +1$ to halve the work — and $\varepsilon \mapsto -\varepsilon$ is not a symmetry of the problem, it exchanges a polytope for a different region. With all 2^k sign patterns, and comparing the itineraries the dynamics actually produces against *all* orderings of the chosen mirrors whose product is scalar rather than only the cyclic ones, the example appears at once.

4.10 A 7-reflective projective billiard in $\mathbb{R}^4$

The threshold $k = n + 3$ is also attained one dimension higher. The following example is deliberately given with rational data, including a rational witness.

Theorem 4.25. *Let $Q_0 = X_0^2 + \cdots + X_4^2$ and let*

$$n_0 = (0,1,0,0,-1),\ n_1 = (0,0,1,0,-1),\ n_2 = (0,0,0,1,0),$$
$$n_3 = (1,-1,0,0,0),\ n_4 = (0,1,1,0,0),\ n_5 = (1,0,0,0,1),$$
$$n_6 = (1,0,-1,0,0).$$

These are roots of B_5. Put $R_i = I - 2n_i n_i^T/(n_i^T n_i)$, let

$$T = \begin{pmatrix} -1/3 & -1/3 & 1/3 & 1/3 & 1/3 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \qquad Q = T^T T,$$

and use the affine chart whose last homogeneous coordinate is 1. The seven walls, oriented inward, are

$$\begin{aligned} -x + w &< 0, & -y + w &< 0, \\ z &< 0, & 4x + y - z - w - 1 &< 0, \\ -x - y &< 0, & x + y - z - 4w - 1 &< 0, \\ & & x + 4y - z - w - 1 &< 0. \end{aligned}$$

Then their projective billiard, with the polar fields for Q, is a bounded convex 7-reflective projective billiard of primitive period 7. A rational witness starts on the fifth wall at

$$p_0 = (0, 0, -4/7, -2/21), \qquad u_0 = (1/5, 1, -1/4, -1/5),$$

and has successive first impacts on the walls

$$5, 6, 0, 1, 2, 3, 4.$$

It closes with $p_7 = p_0$ *and* $u_7 = u_0$.

Proof. The explicit B_5 calculation gives

$$R_6R_5R_4R_3R_2R_1R_0 = -I_5.$$

Thus $H_i = T^{-1}R_iT$ are Q-reflections and their product is projectively scalar. The displayed affine inequalities are precisely the oriented images of the $n_i^{\perp}$, cleared of denominators so as to be legible. The positive dependence is in the *unscaled* normalisation: taking the covectors $n_i^{\top}T$ with the sign that makes the interior negative, their seven spatial parts sum to zero, so $\mu = (1, \dots, 1)$ is a strictly positive dependence and the recession cone is trivial; those seven vectors have rank four. Clearing denominators rescales three of the seven and destroys the all-ones dependence, so the certificate must be read before that step. Finally $(1/10, 1/10, -1/10, 0)$ is strictly interior. So the domain is nonempty, bounded and convex.

For the stated (p_0, u_0), exact rational ray tracing gives the seven positive flight times

$$\frac{4}{189}, \frac{74}{867}, \frac{1050}{10693}, \frac{105}{703}, \frac{1575}{49799}, \frac{17340}{112703}, \frac{20}{129}.$$

At each step the indicated wall is the unique first wall and the impact is strictly interior relative to it. Applying H_i to the homogeneous tangent vector and returning to the affine chart gives $p_7 = p_0$ and $u_7 = u_0$ exactly. The strict inequalities persist under perturbation of the initial data, so Theorem 3.3 gives an open set of 7-periodic orbits. Since the seven walls are distinct, the exhibited period is primitive. □

Remark 4.26 (Reproducibility; calculation added by Codex, 8 August 2026). Every equality and strict inequality in Theorem 4.25 is checked over $\mathbb{Q}$ by `verificadores/reflectivo_rp4_b5.py`. In particular, the theorem does not rely on the floating-point search that located the word.

4.11 An 8-reflective projective billiard in $\mathbb{R}^5$

The preceding word extends once more, and in this case the extension is also realised dynamically.

Theorem 4.27. *There is a bounded convex* 8*-reflective projective billiard in* $\mathbb{R}^5$ *with primitive period* 8*, preserving a positive definite quadric. It may be chosen with rational data. More precisely, take in* $\mathbb{R}^6$ *the roots*

$$(0,1,0,0,-1,0), (0,0,1,0,-1,0), (0,0,0,1,0,0), (1,-1,0,0,0,0),$$
$$(0,1,1,0,0,0), (1,0,0,0,1,0), (1,0,-1,0,0,0), (0,0,0,0,0,1),$$

with orientations $(+,+,-,+,-,-,+,-)$ *and the affine chart defined by* $\ell = (-1,1,1,1,3,1)$. *The associated* Q*-reflections have product* $-I_6$. *The point and direction*

$$p_0 = (-2/33, 0, 0, 4/11, 4/11), \qquad u_0 = (-11/10, 1, -2/5, 7/10, 1/5)$$

start on wall 4 *and have first-impact itinerary*

$$5, 6, 7, 0, 1, 2, 3, 4,$$

closing exactly with $p_8 = p_0$ *and* $u_8 = u_0$.

Proof. The first seven roots are those of Theorem 4.25, extended by zero, and their product is $\mathrm{diag}(-I_5, 1)$. The final coordinate reflection makes the product $-I_6$. In the stated chart the eight inward spatial normals have positive dependence with all coefficients 1, and the point $(-1/10, 1/10, 1/10, 1/10, 1/10)$ is strictly interior, proving boundedness and nonemptiness. Exact rational tracing from the displayed data has the stated itinerary, with every competing impact strictly later and every impact interior relative to its wall; it returns exactly to (p_0, u_0). Theorem 3.3 then supplies an open set of 8-periodic trajectories. □

Remark 4.28 (Reproducibility; calculation added by Codex, 8 August 2026). The full list of affine inequalities, flight times and strict comparisons for Theorem 4.27 is checked over $\mathbb{Q}$ by `verificadores/reflectivo_rp5_b6.py`.

4.12 A 9-reflective projective billiard in $\mathbb{R}^6$

The threshold is attained once more, two dimensions higher, and with k odd.

Theorem 4.29. *Let $Q = \sum_{j=1}^{7} x_j^2$ and let the nine mirrors of $\mathbb{RP}^6$ have normals*

$$\begin{aligned} &n_0 = (0,0,0,0,0,1,-1), \quad n_1 = (0,1,0,1,0,0,0), \quad n_2 = (0,0,1,0,0,0,0),\\ &n_3 = (0,0,0,1,0,1,0), \quad n_4 = (1,0,0,0,0,0,0), \quad n_5 = (0,1,0,0,0,0,-1),\\ &n_6 = (0,0,0,0,1,0,0), \quad n_7 = (0,0,0,1,0,0,-1), \quad n_8 = (0,1,0,0,0,1,0), \end{aligned}$$

with transversal field on face i the pencil through the pole $O_i = n_i$ of Q, and take the affine chart $\ell \cdot X = 1$ with $\ell = (-1,1,-1,-3,1,1,-1)$, solving for the fourth homogeneous coordinate, and the sign vector $\varepsilon = (-,+,+,+,+,-,-,+,-)$. The resulting bounded convex polytope of $\mathbb{R}^6$ is a 9-reflective projective billiard.

Proof. The nine reflections are involutions of $\mathrm{O}(Q)$ with $R_8 \cdots R_0 = -I_7$, which is (H1). The polytope is bounded because the nine outward normals in the chart *sum to zero* — so $\mu = (1, \dots, 1)$ is a strictly positive certificate — and have rank six. The point

$$p_0 = (-\tfrac{6}{19}, \tfrac{7}{228}, -\tfrac{6}{19}, 0, \tfrac{7}{228}, -\tfrac{7}{228})$$

lies on face 6 and strictly inside the other eight. Exact rational tracing from p_0 in the direction

$$u_0 = 10^{-6}(-1003425, -270995, 31421, 1824019, -819948, -545098)$$

gives the itinerary $7, 8, 0, 1, 2, 3, 4, 5, 6$, with every impact strictly interior to its face, every competing face excluded by a strict inequality, all flight parameters positive, and a return to p_0 with $\kappa = 1$. Theorem 3.3 applies. □

Remark 4.30. Three points. First, $k = 9$ is odd and $n = 6$ is even, which is exactly what Theorem 4.6 permits; in odd ambient dimension k would have been forced even. Second, $9 = n+3$, so this is again the threshold, as in Theorems 4.13, 4.23, 4.33, 4.25 and 4.27. Third, and this is the informative part, the word obeys Theorem 4.31 on the nose: six of its roots are of type $e_i \pm e_j$ and live on the coordinates $\{1, 3, 5, 6\}$, forming a D_4 block of rank four, and the other three are the sign changes e_0, e_2, e_4. The block has the same multiset of $\cos^2$ of pairwise angles as the cores of Theorems 4.25 and 4.27 and as the configuration of Theorem 4.33: it is the same rank-four object again, and the three isolated coordinates make the configuration reducible, of corank three. Checked over $\mathbb{Q}$ by `verificadores/reflectivo_rp6_b7.py`.

4.13 Why the B_m route stops at rank four

The words used in Theorems 4.25 and 4.27 live in B_5 and B_6, and it is natural to ask how far that supply goes. Not far, and the reason is a counting argument inside S_m.

Theorem 4.31 (The B_m threshold words are reducible for $m \geq 5$)**.** *Let $m \geq 5$ and let $r_1 \cdots r_{m+2} = -I_m$ be a word of $m+2$ reflections of B_m. Then the configuration is reducible: some proper nonzero subspace of $\mathbb{R}^m$ is invariant under all the r_i. The same holds in D_m.*

Proof. Write $B_m = (\mathbb{Z}/2)^m \rtimes S_m$. A reflection in e_i has trivial permutation part; a reflection in $e_i \pm e_j$ has permutation part the transposition (ij). Since $-I_m$ has trivial permutation part, the t transpositions occurring must multiply to the identity of S_m.

If some coordinate c occurs in no root of type $e_i \pm e_j$ then $\langle e_c \rangle$ is invariant under every r_i; and if the graph whose edges are those transpositions is disconnected, the span of the coordinates of one component is invariant. So irreducibility forces that graph to be connected and to span all m vertices, that is, the transpositions act transitively.

A transitive factorisation of the identity of S_m into t transpositions satisfies $t = 2(m-1) + 2g$ with $g \geq 0$, hence $t \geq 2(m-1)$; and $t \leq m+2$. Therefore $2(m-1) \leq m+2$, that is $m \leq 4$. In D_m every root is of type $e_i \pm e_j$, so $t = m+2$ and the same inequality applies. □

Corollary 4.32. *Let $m \geq 5$ and let a word of $m+2$ reflections with product $-I_m$ generate a finite reflection group acting irreducibly on $\mathbb{R}^m$. Then that group is E_7 (with $m = 7$) or E_8 (with $m = 8$).*

Proof. The group has rank m, acts irreducibly and contains $-I_m$, which is the product. Among the irreducible finite reflection groups containing $-I$, those of rank at least five are B_m, D_m with m even, E_7 and E_8; Theorem 4.31 removes the first two. □

The inequality is sharp and says more than the statement. At $m = 4$ it forces $t = 6$ and no sign changes at all, so every irreducible threshold word of rank four consists of six roots of type $e_i \pm e_j$: a D_4 word. Deleting the isolated coordinates from the words of Theorems 4.25 and 4.27 leaves exactly such a core, and it is the same core in both. The uniform supply of $m+2$ reflections of B_m with product $-I_m$, for every $m \geq 5$, is therefore that one rank-four configuration padded with coordinate reflections, and its algebraic content does not grow with m. That the core is more than algebra is the next example.

Theorem 4.33. *The bounded convex polytope of $\mathbb{R}^3$ cut out by the six faces with normals*

$$n_0 = (0,1,0,-1), \quad n_1 = (0,0,1,-1), \quad n_2 = (1,-1,0,0),$$

$$n_3 = (0,1,1,0), \quad n_4 = (1,0,0,1), \quad n_5 = (1,0,-1,0)$$

in the affine chart $\ell = (-2,-2,-2,-1)$, with transversal field on face i the pencil through the pole $O_i = n_i$ of $Q = \sum_j x_j^2$, is a 6-reflective projective billiard. It is not projectively equivalent to the billiard of Theorem 4.23.

Proof. The six reflections satisfy $R_5 \cdots R_0 = -I_4$, which is (H1). The polytope is bounded, by an explicit positive rational combination of the outward normals summing to zero together with those normals having rank three, and nonempty, the point $(-\frac{1}{9}, -\frac{1}{9}, \frac{1}{9})$ being strictly interior to all six half-spaces. The ray from that point in direction $(\frac{5}{4}, -\frac{1}{4}, 2)$ meets face 3 first; starting there, exact rational tracing gives the itinerary $4, 5, 0, 1, 2, 3$ with every impact strictly interior to its face, every competing face excluded by a strict inequality, all flight parameters positive, and a return to the starting point with $\kappa = 1$. Theorem 3.3 applies. For the last claim, both configurations are irreducible, so each preserves a unique quadric up

to scale; a projective equivalence between them would carry Q to a multiple of Q and hence be orthogonal up to scale, but the multiset of $\cos^2$ of the angles between normals has three zero entries here and six for Theorem 4.23. □

Remark 4.34. Its mirror group is all of $W(D_4)$, of order 192, exactly as for Theorem 4.23. So this is a second kaleidoscope in $\mathbb{R}^3$ rather than a third kind of example, and like the symmetric pentagon of the plane it unfolds to a finite group and has *every* orbit periodic. The example with infinite mirror group in $\mathbb{R}^3$ remains the one of Theorem 4.13. Checked over $\mathbb{Q}$ by `verificadores/reflectivo_rp3_d4b.py`.

4.14 A rank bound, and a lower bound on k

Before returning to irreducibility, one lemma that costs a line and constrains everything.

Lemma 4.35 (Rank bound)**.** *For the canonical lifts $H_i = I - 2O_iL_i/(L_iO_i)$ of harmonic homologies of* PGL_m*,*

$$\operatorname{rank}\big(H_{j-1}\cdots H_0 - I\big) \le j.$$

Proof. The product telescopes: $H_{j-1}\cdots H_0 - I = \sum_{i=0}^{j-1} H_{j-1}\cdots H_{i+1}(H_i - I)$, a sum of j matrices each of rank 1, since $H_i - I = -2O_iL_i/(L_iO_i)$. □

Corollary 4.36. (a) *A product of $j < m$ harmonic homologies has eigenvalue* 1 *with multiplicity at least $m - j$. In particular a product of three of them in* PGL_4 *always has eigenvalue* 1*.*

(b) *If $H_{k-1}\cdots H_0 = cI$ with $c \neq 1$, then* $\operatorname{rank}(cI - I) = m \le k$*, i.e.*

$$\boxed{k \ge n+1}$$

There are no k-reflective projective billiards of $\mathbb{RP}^n$ with $k < n+1$ and nontrivial scalar. For $n = 2$ this is $k \ge 3$, which is sharp: Fierobe's triangle.

(c) *Consequently, in the splitting $6 = 3+3$ of Remark 4.22 the condition* $\det(X - I) = 0$ *is one genuine condition when $c = -1$ — eigenvalue* 1 *comes free by* (a)*, and eigenvalue* -1 *does not — and is automatic when $c = +1$.*

Proof. (a) is Lemma 4.35 read as $\dim\ker(M - I) \ge m - j$. (b) is the same lemma applied to the product, since $\operatorname{rank}(cI - I) = m$ for $c \neq 1$. (c) follows from (a) with $j = 3$, $m = 4$, plus the computation $\det(X - I) = \pm\det(M - cI)$ for $X = cM^{-1}$. □

4.15 A conditional reduction for the regular part of V_6

Use determinant-one matrix lifts and work over $\mathbb{C}$. Put

$$\mathcal{P} = \{M \in \mathrm{SL}_4 : \operatorname{rank}(M - I) \le 2\}.$$

Every product of two normalized harmonic homologies belongs to $\mathcal{P}$. The distinction between the full determinantal locus and the image of the factorisation map matters at exceptional matrices.

Lemma 4.37 (The pair locus and its regular fibres)**.** *$\mathcal{P}$ is irreducible of dimension* 11*. On its nonempty open subset $\mathcal{P}^{\mathrm{rs}}$ of matrices with eigenvalues $1, 1, \lambda, \lambda^{-1}$, $\lambda \neq 0, \pm 1$, and two-dimensional* 1*-eigenspace, the factorisation map from pairs of harmonic homologies is smooth with geometrically irreducible one-dimensional fibres. The fibre over I instead has dimension* 6*.*

Proof. Consider the incidence space of pairs (F, M) with $F \in \mathrm{Gr}(2,4)$ and $M|_F = I$. In local coordinates for the Grassmannian, such a matrix has blocks $\left(\begin{smallmatrix} P & 0 \\ R & I \end{smallmatrix}\right)$, with $P \in \mathrm{SL}_2$ and arbitrary R. This is a locally trivial irreducible family of dimension $4+3+4 = 11$, surjecting onto $\mathcal{P}$. Over the rank-two open subset $F = \ker(M - I)$ is unique, proving the dimension.

For $M \in \mathcal{P}^{\mathrm{rs}}$, write $F = \ker(M - I)$ and let E be the sum of the λ and λ^{-1} eigenlines. A factor H_0 obeys $H_0 M H_0 = M^{-1}$, so preserves F, E and interchanges those eigenlines. Its restriction to E already has a one-dimensional negative eigenspace; since a harmonic homology has exactly one negative eigenline, it is identity on F. In an eigenbasis of E it is $\left(\begin{smallmatrix} 0 & q \\ q^{-1} & 0 \end{smallmatrix}\right)$, $q \neq 0$. Conversely each such choice gives the pair $H_1 = MH_0, H_0$. The fibre is $\mathbb{G}_m$. Ordering the simple eigenvalues and choosing eigenvectors etale locally proves smoothness. For $M = I$ one simply has $H_1 = H_0$, and all homologies are allowed, a six-dimensional family. □

Group the six mirrors into $A = H_1H_0$, $B = H_3H_2$, $C = H_5H_4$. Scalar monodromy forces $C = c(BA)^{-1}$ and $\mathrm{rank}(BA - cI) \leq 2$. Define

$$\mathcal{B}_c = \{(A, B) \in \mathcal{P}^2 : \mathrm{rank}(BA - cI) \leq 2\},$$

and let $\mathcal{B}_c^{\mathrm{rs}}$ impose in addition $A, B, c(BA)^{-1} \in \mathcal{P}^{\mathrm{rs}}$. Above this open subset the six-mirror locus is a smooth family of geometrically irreducible fibres of dimension 3.

Proposition 4.38 (Conditional global reduction)**.** *Suppose $\mathcal{B}_c^{\mathrm{rs}}$ is nonempty and irreducible and every component of V_6 meets its inverse image. Then V_6 is irreducible; under the hypotheses of Corollary 4.18 this implies its threshold conclusion. Neither hypothesis on the global components is established here.*

Proof. The smooth factorisation map over the indicated irreducible base has geometrically irreducible fibres. Smoothness makes it open, so all components dominate the base, and irreducibility of the geometric generic fibre leaves exactly one. By the assumed intersection with every component, this open subset is dense in all of V_6; its closure is therefore all of V_6. □

The expression $11 + 11 - 4 + 3 = 21$ is an expected dimension count, not a proof that the pulled-back rank condition has codimension four. The dimension 21 is known at the submersive examples by the independent differential calculation. Exceptional pair fibres and possible additional components must be controlled before promoting the regular reduction to a global one.

Finally, a local parameterisation of harmonic homologies will be useful.

Lemma 4.39 (Rank-one chart for harmonic homologies)**.** *For a real vector space V of dimension m, every projective harmonic homology has a lift*

$$N = 2w\tilde{w}/(\tilde{w}w) - I,$$

where $w \in V$, $\tilde{w} \in V^*$ and $\tilde{w}w \neq 0$. This identifies the space of such homologies with the open subset $\tilde{w}w \neq 0$ of $\mathbb{P}(V) \times \mathbb{P}(V^*)$, which is irreducible of dimension $2m - 2$.

Proof. The displayed map fixes w and acts by $-I$ on $\ker \tilde{w}$, so it represents a homology; changing its sign gives the usual involutive lift that fixes its axis. Conversely, the centre line and axis hyperplane of any harmonic homology give w and $\ker \tilde{w}$, and their non-incidence is exactly $\tilde{w}w \neq 0$. The parameter space is a nonempty open subset of a product of projective spaces. □

Proposition 4.40 (The determinantal locus is a $(1,1)$-hypersurface)**.** *Let $Y \in \mathrm{PGL}_4$ with $\operatorname{rank}(Y-I)=3$, write $C=Y-I$, let κ span $\ker C$ and ℓ span $\ker C^\top$. In the parameterisation of Lemma 4.39, the locus $\mathcal{D}_Y=\{H\in \mathrm{Inv} : \operatorname{rank}(HY-I)\le 2\}$ is cut out by*

$$\text{(i) } \ell w=0, \qquad \text{(ii) } \tilde{w}\kappa=0, \qquad \text{(iii) } 2\,b(\tilde{w})^\top w=\tilde{w}w,$$

where $b(\tilde{w})$ solves $b^\top C=\tilde{w}Y$ and depends linearly on $\tilde{w}$. Conditions (i) *and* (ii) *are linear, so $\mathcal{D}_Y$ is a* ***bidegree*** $(1,1)$ ***hypersurface in*** $\mathbb{RP}^2\times\mathbb{RP}^2$*, irreducible of dimension* 3 *whenever its coefficient matrix has rank ≥ 2.*

Proof. Writing $H=I-2w\tilde{w}/s$ with $s=\tilde{w}w$, we get $HY-I=C-\frac{2}{s}\,w\,(\tilde{w}Y)$, a rank-one perturbation of C. By Wedderburn's rank-one reduction, if $u=Ca$ and $v^\top=b^\top C$ then $\operatorname{rank}(C-t\,uv^\top)=\operatorname{rank}C-1$ exactly when $t\,b^\top Ca=1$; and the perturbation lowers the rank at all only if $u\in\operatorname{im}C$ and $v^\top\in\operatorname{im}C^\top$. The first is $\ell w=0$. For the second, $\tilde{w}Y$ must annihilate $\ker C$, and $Y\kappa=\kappa$ gives $\tilde{w}Y\kappa=\tilde{w}\kappa$, which is (ii). The scalar condition is $\frac{2}{s}\,b^\top w=1$, i.e. (iii), and $b^\top w$ is well defined because b is determined modulo ℓ and $\ell w=0$. Bidegree $(1,1)$ forms are reducible only if they factor as a form in w times a form in $\tilde{w}$, i.e. only in rank 1. □

Remark 4.41. At the configuration of Theorem 4.13 this is checked in exact arithmetic: $\operatorname{rank}(X-I)=3$, both kernels are lines, and the 3×3 coefficient matrix of (iii) is

$$\begin{pmatrix} 0 & -\frac{65}{24} & -\frac{11}{24} \\ \frac{65}{24} & 0 & -\frac{7}{12} \\ \frac{11}{24} & \frac{7}{12} & 0 \end{pmatrix},$$

antisymmetric of rank 2: so $\mathcal{D}_X$ is irreducible there. What is left for the threshold theorem in dimension three includes irreducibility of the regular base and control of the exceptional components, as made explicit in Proposition 4.38.

4.16 Which values of k admit a metric example

Existence and *metric* existence are different questions, and it is the second one that has structure. Call a k-reflective projective billiard *metric* if it preserves a definite conic; by Proposition 2.2 that is the same as being the metric billiard of the corresponding round projective metric; its spherical lift is round by the positive definite Cayley–Klein construction.

Three facts, of which the first two are new here and the third is Theorem 4.9 read correctly.

Proposition 4.42 ($k=m$: metric, and unique)**.** *Let $k=m=n+1$, suppose the configuration is metric, and suppose its lifted product is $-I_m$. Then the normals are pairwise orthogonal, so the configuration is the orthant bounded by m mutually orthogonal mirrors, unique up to isometry. It is realised for every n.*

Proof. Conjugate the definite form to the Euclidean inner product and argue by induction on m, with $m=1$ immediate. Multiplication of the product relation on the right by H_0 gives $H_{m-1}\cdots H_1=-H_0$. Telescoping shows

$$\operatorname{im}(-H_0-I)\subset\operatorname{span}(n_1,\dots,n_{m-1}).$$

For this inclusion, use $A_r\cdots A_1-I=(A_r-I)A_{r-1}\cdots A_1+(A_{r-1}\cdots A_1-I)$, so every summand has image in one of the displayed normal lines. The left side equals $n_0^\perp$ and has dimension $m-1$, so the two spaces are equal. Thus every other normal is orthogonal to n_0. Restriction to $n_0^\perp$ gives a product of $m-1$ reflections equal to $-I$ there; the induction

hypothesis finishes the proof. For the realisation take $H_i = I - 2e_i e_i^*$ on the simplex $\{x_i > 0\}$, with transversal field on the facet $x_i = 0$ the pencil through the opposite vertex $[e_i]$: the homologies commute, their product is $-I$, and the line $f(s,t) = (s - \lambda_0 t, \dots, s - \lambda_n t)$ with $\lambda_0 > \cdots > \lambda_n > 0$ crosses the facets one at a time, in strict order, with all other coordinates on the correct side of the orthant after folding. □

For $n = 2$ this is the classification of 3-reflective spherical billiards by three orthogonal great circles [2]; the proposition is its extension to every dimension, and the point is not the construction but the uniqueness, which comes for free from saturation of the rank bound.

Proposition 4.43 (Common centre kills every nondegenerate conic)**.** *A configuration of harmonic homologies with a common centre O and at least two distinct axes preserves no nondegenerate quadric.*

Proof. If S is invariant and nondegenerate, each H_i is an involution preserving S, so its $+1$ and -1 eigenspaces are S-orthogonal. The -1 eigenspace is $\langle O \rangle$ and the $+1$ eigenspace is $\ker L_i$, so $SO \in \langle L_i \rangle$. As S is nondegenerate and $O \neq 0$ we have $SO \neq 0$, hence $L_i \propto SO$ for every i and all the axes coincide. □

Corollary 4.44 (The rigid end is not empty; it is empty of metric billiards)**.** *Theorem 4.9 says there is no metric k-reflective billiard with $k = m+1$. It does not say the value $k = m+1$ is unattainable: centrally projective configurations, all of whose centres coincide, realise it, and by Proposition 4.43 they preserve no nondegenerate conic at all. So at the rigid end the absence of an invariant conic is not a choice of construction: it is forced.*

Remark 4.45 (The metric table)*.* Collecting, with $m = n + 1$:

$k = m$	metric with product $-I$, and then unique	Proposition 4.42
$k = m + 1$	never metric	Theorem 4.9, Corollary 4.44
$k = m + 2$	metric examples exist	Section 4
$k \geq m + 3$	not classified here	

The determinant condition for a metric configuration with product $-I$ is $k \equiv m \pmod 2$. Beyond the values proved in this paper, the existence of metric examples is left open here. The displayed finite computations are checked over $\mathbb{Q}$ by `verificadores/simplex_ortogonal.py` and `verificadores/extremo_rigido_no_metrico.py`.

4.17 Why four, in any simply-laced system

Theorem 4.31 was proved by counting transitive factorisations in S_m, which is unavailable outside B_m and D_m. The bound has a shorter and more general reason, and it explains where the number four comes from.

Lemma 4.46 (Four blocks at most)**.** *Let $\{r_1, \dots, r_m\}$ be an orthogonal set of roots of a simply-laced system, normalised to $|r_i|^2 = 2$, and let a be another root. Then $\sum_i \langle a, r_i \rangle^2 = 4$, and since the lattice is integral at most **four** of the products are nonzero.*

Proof. Write $a = \sum c_i r_i$ with $c_i = \langle a, r_i \rangle / 2$. Then $2 = |a|^2 = \sum c_i^2 |r_i|^2 = \frac{1}{2} \sum \langle a, r_i \rangle^2$. □

Corollary 4.47. *If a reflection group is generated by an orthogonal set of m roots together with one further root, the number of irreducible blocks satisfies $s \geq m-3$; so it acts irreducibly only if $m \leq 4$.*

The count is sharp and visible in the root systems: with respect to a maximal orthogonal set, no root of E_7 or of E_8 meets all the blocks, while in D_4 sixteen of the twenty-four roots do. That is why rank four is the last one where an irreducible threshold word can occur, and it is the same four as in Theorem 4.31 — there it looked like a fact about signed permutations, and it is really a fact about $|r|^2 = 2$. Checked over $\mathbb{Z}$ by `verificadores/cuatro_bloques.py`.

4.18 The equation $\det(N - A) = 0$, in any dimension — and why it stops

In vector rank three, $\det(N - A) = 0$ becomes $bw_2 = cw_3$ in an eigenbasis of A, a bidegree $(1, 1)$ equation with coefficient matrix of rank 2. The general formula below shows why this condition becomes identically zero in vector rank four.

Lemma 4.48. *Let $A \in \mathrm{PGL}_m$, $c = \pm 1$, and $N = c\big(2w\tilde{w}/s - I\big)$ with $s = \tilde{w}w$. Put $C = cI + A$. Then*

$$\det(N - A) = 0 \iff \tilde{w}\, E_A\, w = 0, \qquad \boxed{E_A := \det(C)\, I - 2c\,\mathrm{adj}(C)}$$

So $\mathcal{C}_A = \{N : \det(N-A) = 0\}$ is a bidegree $(1, 1)$ hypersurface in $\mathbb{RP}^{m-1} \times \mathbb{RP}^{m-1}$, irreducible of dimension $2(m-1)-1$ as soon as $\operatorname{rank} E_A \geq 2$. When C is invertible, $\operatorname{adj} C = \det(C)C^{-1}$, so $E_A = \det(C)\big(I - 2cC^{-1}\big)$ and $\operatorname{rank} E_A = \operatorname{rank}(A - cI)$.

Proof. $N - A = -C + \frac{2c}{s} w\tilde{w}$, and $\det(C - t\, w\tilde{w}) = \det C - t\, \tilde{w}\,\mathrm{adj}(C)w$ for any C, invertible or not. Multiply by s. □

Corollary 4.49. *For $m = 3$ and $c = +1$ — the case of V_5 — with A regular semisimple of eigenvalues $1, \lambda, \lambda^{-1}$, $C = I + A$ is invertible and in that eigenbasis E_A is proportional to $\mathrm{diag}(0, 1, -1)$, so $\tilde{w}E_A w = 0$ is $bw_2 = cw_3$ and $\operatorname{rank} E_A = 2$. The involution-chart lemma of This recovers the rank-three equation directly.*

Proposition 4.50 (Why it stops at $m = 4$)**.** *For $m = 4$ and $c = -1$ — the case of V_6 — one has $C = A - I$, and $A \in \mathcal{P}$ forces $\operatorname{rank}(A - I) \leq 2 = m - 2$. The adjugate of a matrix of corank ≥ 2 vanishes, and so does its determinant. Hence*

$$E_A = 0 \quad \textit{identically,}$$

*and the condition $\det(N - A) = 0$ is **automatic**: it carries no information.*

Remark 4.51 (Where this leaves the threshold in dimension three). Consequently the fibration modelled on the V_5 proof does not close. Its stages would be $A \in \mathcal{P}$ (11), $N \in c\,\mathrm{Inv}$ with no condition by Proposition 4.50 (6), $H_4 \in \mathcal{D}_Y$ (3 by Proposition 4.40), and the two factorisations $(1 + 1)$: total 22, whereas the submersive examples have local dimension 21. This parameter count therefore cannot describe a neighbourhood of those examples with independent parameters of the stated dimensions. The missing compatibility or fibre constraint is *not* $\det(N - A) = 0$; identifying it, and controlling exceptional components, remain open.

So: irreducibility of V_6 is **not** proved, and neither is the threshold theorem in dimension three. What is proved unconditionally is Proposition 4.12: the component of V_6 through the explicit rational point is contained in S_6, so every configuration on *that* component preserves a quadric. What is new here is the dimension-free Lemma 4.48, which explains where $bw_2 = cw_3$ came from, and Proposition 4.50, which turns "the mechanism does not seem to generalise" into a theorem about why: in $\mathbb{RP}^3$ the matrix $A - I$ is forced to have corank two, and adjugates of such matrices vanish.

Reproducibility and computational disclosure

Every displayed rational polytope has an accompanying exact-arithmetic certificate for scalar monodromy, boundedness, genuine facets, strict first impacts, closure and primitivity. Floating-point exploration is not used in any proof. Research and manuscript development for all three papers used AI assistance from early July through late August 2026 (approximate). The models used were OpenAI's GPT-6 Astra, GPT-5.6 Sol, and GPT-5.6 Luna, and Anthropic's Fable 5 and Opus 5, accessed through ChatGPT Work, ChatGPT Codex, and Claude Code. No API was used. The assistance supported research, manuscript drafting, development and checking of verification scripts, and revision. Jorge Lucas González reviewed the manuscripts and assumes full responsibility; no AI system is an author.

Conclusion

The paper establishes the parity obstruction, the general rank lower bound, and a collection of exact higher-dimensional polytope constructions. It also separates the proved threshold statements from the irreducibility problem that remains open in projective three-space. The examples and their exact certificates are therefore reusable independently of that unresolved step.

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References

[1] M. Bialy, C. Fierobe, A. Glutsyuk, M. Levi, M. Plakhov and S. Tabachnikov, *Open Problems on Billiards and Geometric Optics*, Arnold Math. J. **8**(3–4) (2022), 411–422, doi:10.1007/s40598-022-00198-y.

[2] V. Blumen, K. Y. Kim, J. Nance and V. Zharnitsky, *Three-period orbits in billiards on the surfaces of constant curvature*, Int. Math. Res. Not. (2012), 5014–5024, doi:10.1093/imrn/rnr228.

[3] W. Crawley-Boevey, *Quiver algebras, weighted projective lines, and the Deligne–Simpson problem*, Proc. ICM Madrid 2006, vol. II, 117–129.

[4] C. Fierobe, *Examples of projective billiards with open sets of periodic orbits*, Discrete Contin. Dyn. Syst. **44** (2024), 3287–3301, doi:10.3934/dcds.2024059.

[5] C. Fierobe, *On projective billiards with open subsets of triangular orbits*, Israel J. Math. **267** (2025), 501–548, doi:10.1007/s11856-024-02712-z; arXiv:2005.02012.

[6] C. Fierobe, *Only quadrics have pseudo-caustics — on caustics of Riemannian, pseudo-Euclidean and projective billiards in higher dimensions*, Math. Z. **312** (2026), Art. 67, doi:10.1007/s00209-026-03939-3; arXiv:2306.06757.

[7] F. Heidari and B. Honari, *The three harmonic homologies theorem*, Expo. Math. **42** (2024), 125544, doi:10.1016/j.exmath.2024.125544.

[8] S. Tabachnikov, *Introducing projective billiards*, Ergodic Theory Dynam. Systems **17** (1997), 957–976, doi:10.1017/S0143385797086239.